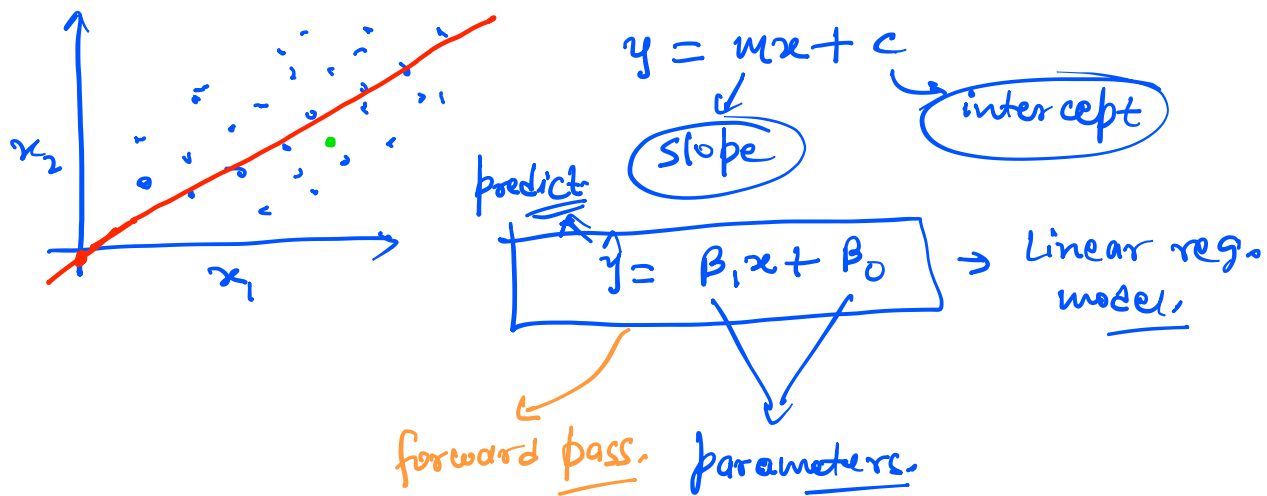


Linear Regression:

Supervised learning. Predicting numeric values.



$$\text{Salary} = \underline{30000} + \underline{8000} \times \text{YOE}$$

$$\text{Salary when } \text{YOE} = 3 \rightarrow 54000$$

Errors:

Ground truth = y		$\bar{y}$ = mean of y
Prediction = $\hat{y}$		$\bar{\hat{y}}$ = mean of $\hat{y}$
n examples (data points)		

$$\textcircled{1} \frac{1}{n} \sum_{i=1}^n (y - \hat{y}) \quad \textcircled{2} (\bar{y} - \bar{\hat{y}})$$

$$\textcircled{3} (\sigma_y - \sigma_{\hat{y}}) \quad \textcircled{4} \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$$

Standard Errors:

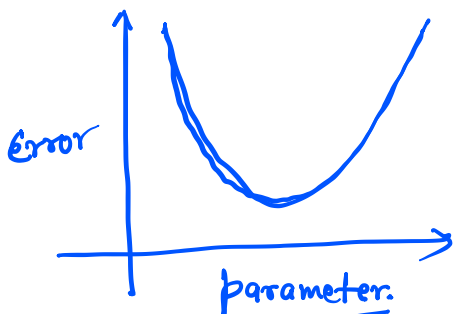
- ① Residual sum of squares:
(RSS) $\sum_{i=1}^n (y_i - \hat{y}_i)^2$
- ② Mean of squared errors:
(MSE) $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- ③ $RMSE = \sqrt{MSE}$
- ④ $MAE = \frac{1}{n} \sum_{i=1}^n (|y_i - \hat{y}_i|)$

y	$\hat{y}$	MSE	MAE	Overall	
				MSE	MAE
100	20	6400	80	3400	50
200	220	400	20		

OLS (Ordinary Least Squares).

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x))^2$$

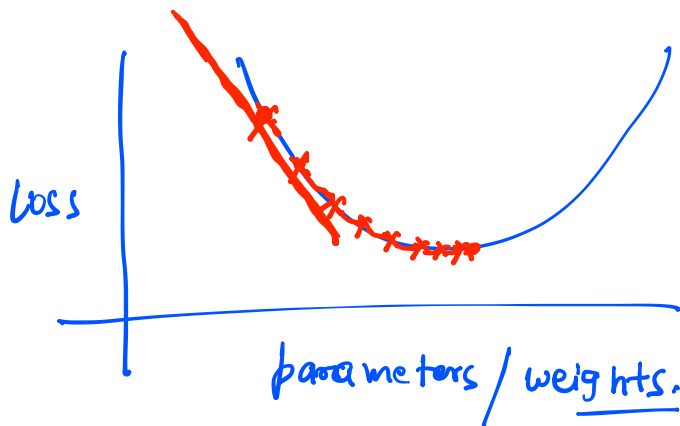


closed form solution:

$$\beta_1 = \frac{\sum_{i=1}^n (\check{x}_i - \bar{x})(\check{y}_i - \bar{y})}{\sum_{i=1}^n (\check{x}_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Gradient descent:



cost/loss
function

$$\hat{y}_i = \beta_1 x_i + \beta_0$$

$$J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{MSE loss})$$

$$J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_1 x_i + \beta_0))^2$$

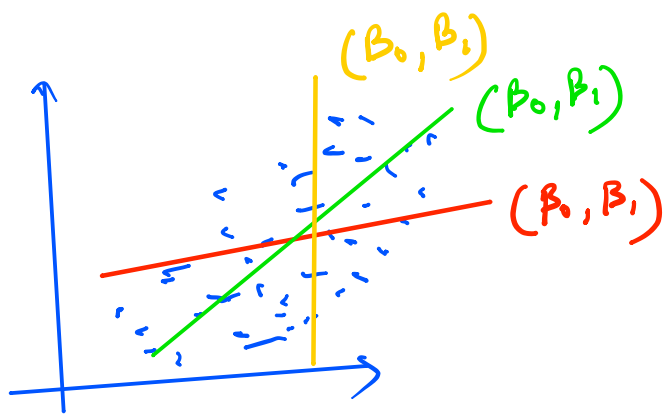
update rule
of grad.
descent.

$$\begin{aligned} \beta_{0_upd} &= \beta_0 - \eta \left(\frac{\partial J}{\partial \beta_0} \right) \\ \beta_{1_upd} &= \beta_1 - \eta \left(\frac{\partial J}{\partial \beta_1} \right) \end{aligned}$$

β_0 and β_1
gets updated for
Each sample.

$\eta \rightarrow$ learning rate (0-1)
hyperparameter.

Epoch $\rightarrow$ no. of
times of iteration
of whole dataset.



$$J(\beta_0, \beta_1) = \frac{1}{2n} \sum_{i=1}^n (y_i - (\beta_1 x_i + \beta_0))^2$$

$$\frac{\partial J}{\partial \beta_1} = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_1 x_i + \beta_0)) \cdot (-x_i)$$

$$\frac{\partial J}{\partial \beta_1} = -\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_i$$

$$\frac{\partial J}{\partial \beta_0} = -\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

Good. Calculation

Good. calculation + weight update

Back propagation

